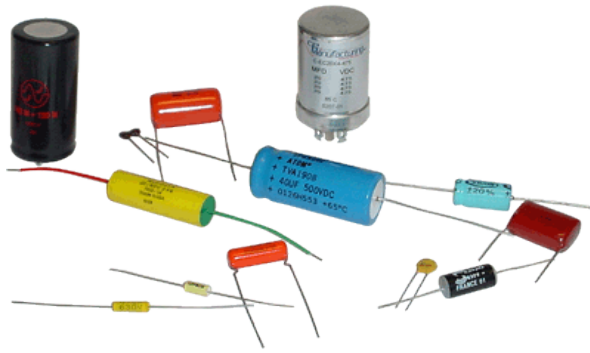




EBN 122

Chapter 6: Capacitors and Inductors

Capacitors and Series and Parallel Capacitors

Chapter 6 overview

6.2. Capacitors

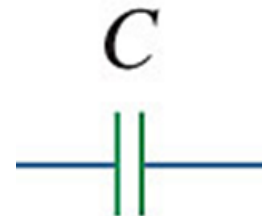
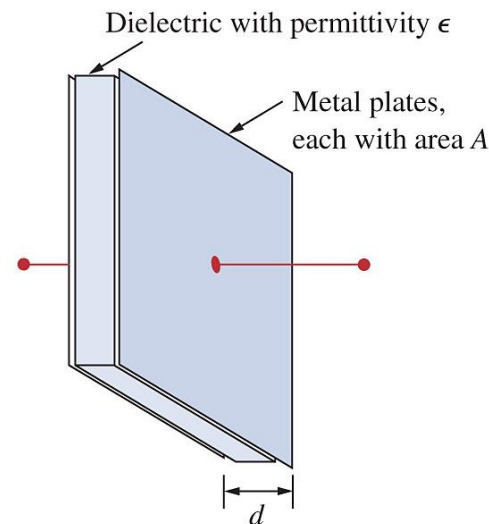
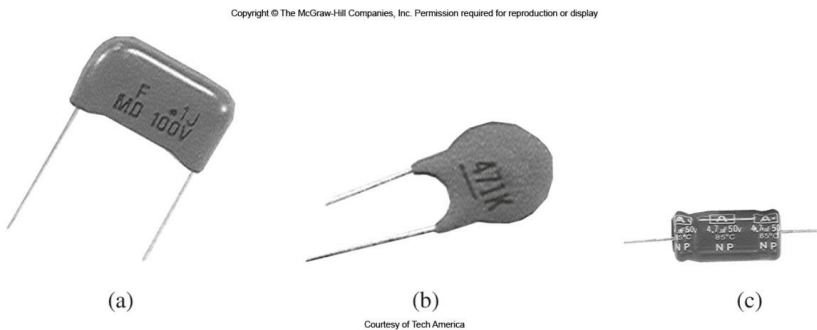
6.3. Series and Parallel
Capacitors

6.4. Inductors

6.5. Series and Parallel
Inductors

Capacitors

- Second most common electrical component after resistors – used extensively
- Passive element that stores energy in its **electric field**
- Consists of 2 conducting plates separated by an insulator or dielectric (air, paper, plastic)

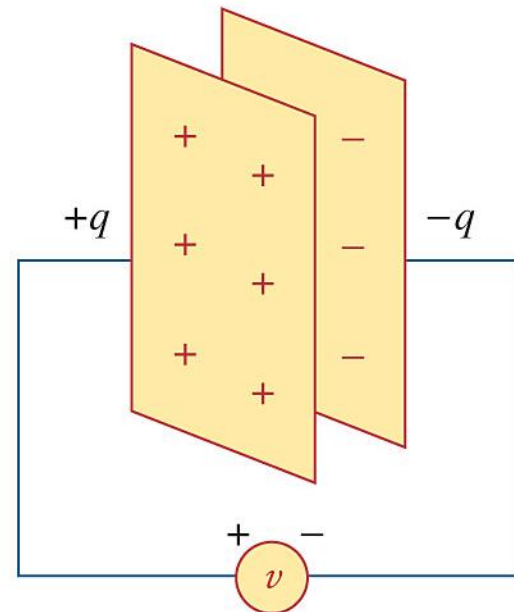


Capacitors

- When voltage source v is connected to the capacitor:
 - positive charge q deposited on one plate
 - negative charge $-q$ deposited on the other
- Capacitor stores the electric charge q , proportional to applied voltage v :

$$q = Cv$$

- **C = capacitance** of capacitor



Capacitors

- Capacitance (C) is measured in farad (F)
- Capacitance = ratio of charge on one plate of capacitor to the voltage difference between the 2 plates:

$$C = \frac{q}{v}$$

- 1 farad = 1 coulomb/volt
- Although C is the ratio of charge per plate (q) to applied voltage (v), it does not depend on q or v

Capacitors

- C is dependent on **physical dimensions** of capacitor:

$$C = \frac{\epsilon A}{d}$$

A = surface area of the metal plate [m²]

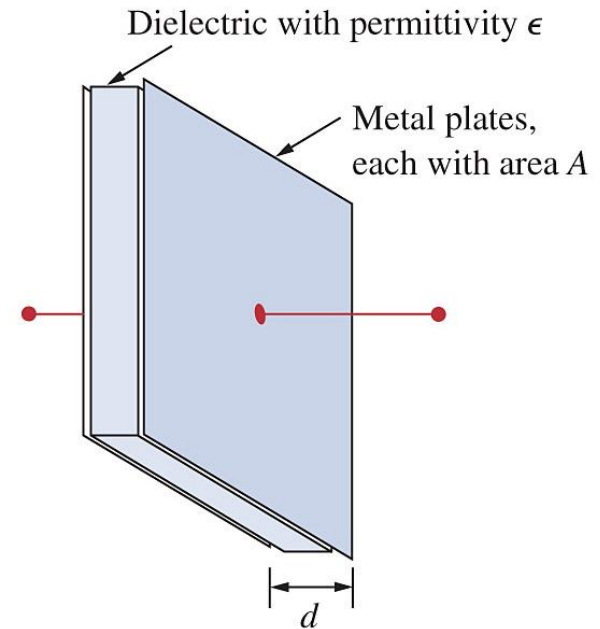
d = distance between plates [m]

ϵ = permittivity of the dielectric [F/m]

$$\epsilon = \epsilon_r \epsilon_0$$

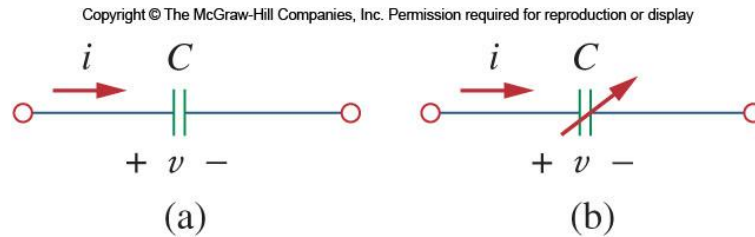
ϵ_r = Relative permittivity of material

ϵ_0 = Permittivity of free space (8.854×10^{-12} F/m)



Capacitors

- Typical values = picofarad (pF) to microfarad (μF)
- Fixed or variable



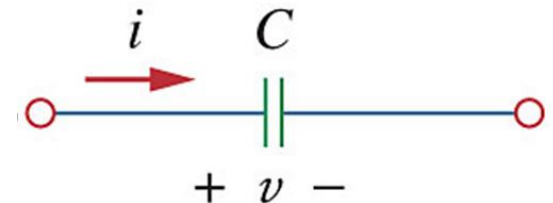
- Different types – polyester, film, electrolytic capacitors

passive sign convention

$v \times i > 0 \rightarrow$ charging

$(v > 0 \text{ and } i > 0, \text{ or } v < 0 \text{ and } i < 0)$

$v \times i < 0 \rightarrow$ discharging



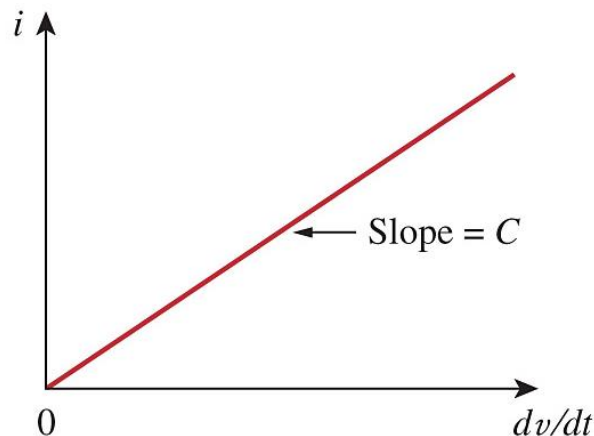
Capacitors

- **Current-voltage** relationship for capacitor:

$$i = \frac{dq}{dt} \quad \text{and} \quad q = Cv$$

Differentiating: $\frac{dq}{dt} = \frac{d}{dt}(Cv)$

$$i = C \frac{dv}{dt}$$



Linear capacitors
→ independent of voltage

Capacitors

- **Voltage-current** relationship for capacitor:

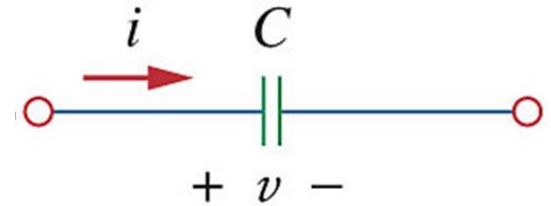
Integrating $i = C \frac{dv}{dt}$

$$v(t) = \frac{1}{C} \int_{-\infty}^t i \, dt = \frac{1}{C} \int_{t_0}^t i \, dt + v(t_0)$$

$$v(t_0) = q(t_0)/C \quad \rightarrow \quad \text{Voltage across capacitor at time } t_0$$

Voltage depends on past history of capacitor current
 $\rightarrow$ capacitor has **memory**

Capacitors



- **Power (p)** delivered to capacitor:

$$p = vi = Cv \frac{dv}{dt} \quad \text{from} \quad i = C \frac{dv}{dt}$$

- **Energy (w)** stored in the capacitor:

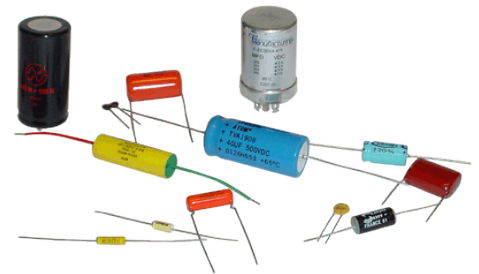
$$w = \frac{1}{2} Cv^2 = \frac{q^2}{2C} \quad \text{from} \quad w = \int_{-\infty}^t p \, dt$$

Capacitor

Important properties of capacitors:

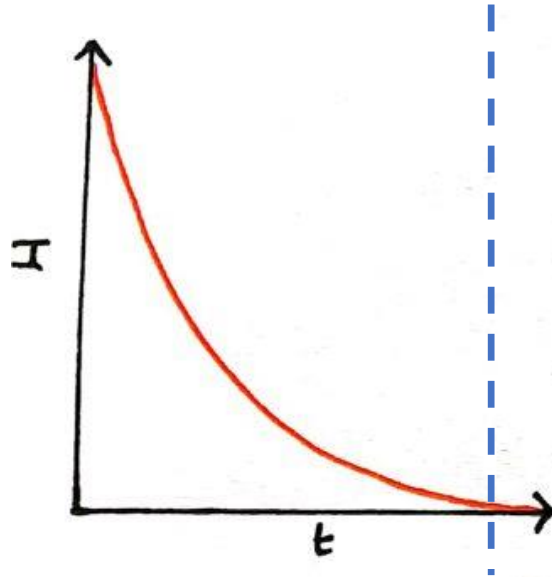
$$i = C \frac{dv}{dt}$$

1. Capacitor is an **open circuit** to dc
(dc $\rightarrow$ voltage not changing $\rightarrow dv/dt = 0$
 $\rightarrow i = 0$, thus open circuit)
2. Voltage on a capacitor **cannot change abruptly**
($dv/dt = \infty \rightarrow i = \infty$, not possible)
3. Does not dissipate energy, but
stores it when **charging**
or **taking** power from circuit (+ p)
or
provides it when **discharging**
or **delivering** power to the circuit (-p)

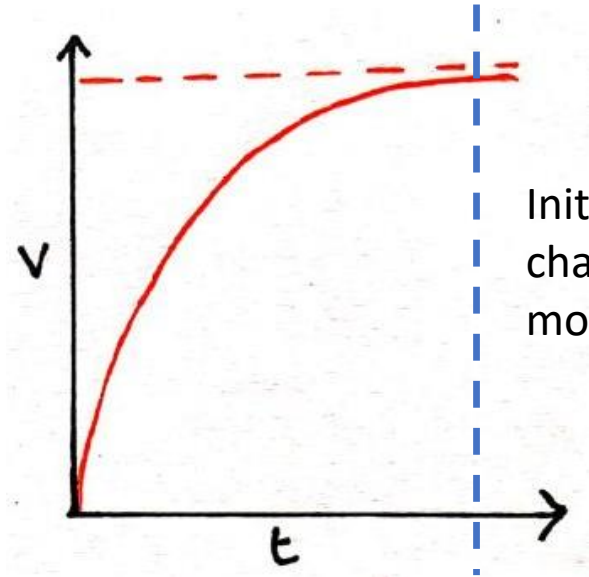


Capacitors

- Current and voltage over time:



DC



Initially, more
charge flows, thus
more voltage

DC

DC-state or steady-state

Capacitors summary

$$q = Cv$$

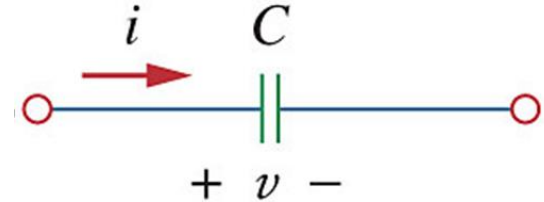
$$C = \frac{\epsilon A}{d}$$

$$i = C \frac{dv}{dt}$$

$$v = \frac{1}{C} \int_{t_0}^t i \, dt + v(t_0)$$

$$p = vi = Cv \frac{dv}{dt}$$

$$w = \frac{1}{2} Cv^2$$



DC conditions:
open circuit

Example

Practice Problem 6.3

If the current through a $100\text{ }\mu\text{F}$ capacitor is given as:

$i(t) = 50 \sin(120\pi t)\text{ mA}$, calculate $v(1\text{ms})$ and $v(5\text{ms})$.

Assume $v(0) = 0\text{ V}$

Example

Practice Problem 6.3

If the current through a $100\text{ }\mu\text{F}$ capacitor is given as:
 $i(t) = 50 \sin(120\pi t)\text{ mA}$, calculate $v(1\text{ms})$ and $v(5\text{ms})$.

Assume $v(0) = 0\text{ V}$

$$v = \frac{1}{C} \int_{t_0}^t i \, dt + v(t_0)$$

$$v(1\text{ms}) = \frac{1}{C} \int_0^{1\text{ms}} 50 \sin(120\pi t) \times 10^{-3} \, dt + 0$$

$$= \frac{-50 \times 10^{-3}}{C \times 120\pi} \cos(120\pi t) \Big|_0^{1\text{ms}}$$

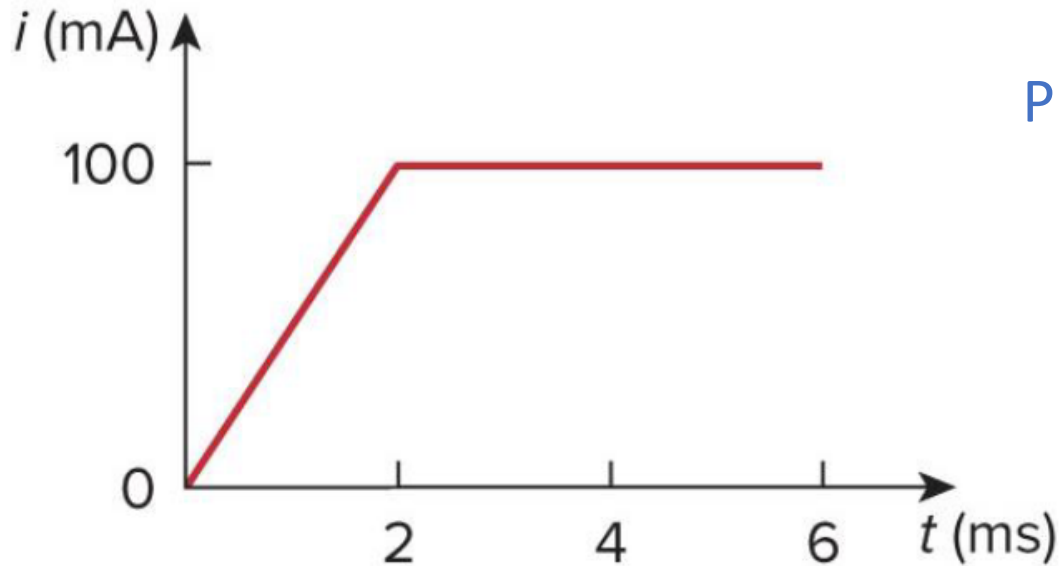
$$= \frac{-50 \times 10^{-3}}{100 \times 10^{-6} \times 120\pi} [\cos(120\pi \times 10^{-3}) - \cos(0)]$$

$$= -1.3263 \times -0.07022 = \underline{93.137\text{ mV}}_{\rightarrow} \text{ at } t = 1\text{ ms}$$

$$v(5\text{ms}) = \frac{1}{C} \int_0^{5\text{ms}} 50 \sin(120\pi t) \times 10^{-3} \, dt + 0 = \underline{1.736\text{ V}}_{\rightarrow}$$

Example 1: Capacitor – voltage and current

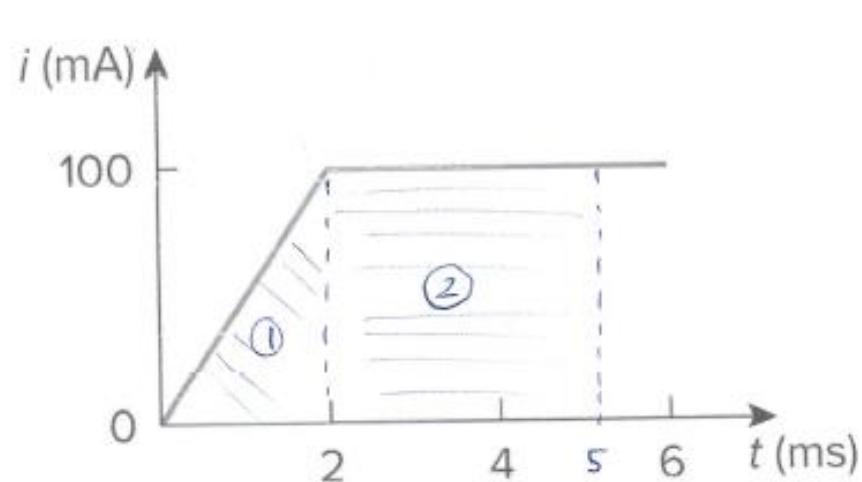
An initially uncharged 1-mF capacitor has the current shown in the figure through it. Calculate the voltage across it at $t=2$ ms and $t=5$ ms.



Practice Problem 6.4

Example 1: Capacitor – voltage and current

An initially uncharged 1-mF capacitor has the current shown in the figure through it. Calculate the voltage across it at $t=2$ ms and $t=5$ ms.



$$v(0) = \frac{1}{C} \int_{-\infty}^0 i(\tau) d\tau = 0$$

$$v(t) = \frac{1}{C} \int_{t_0}^t i(\tau) d\tau + v(t_0)$$

$$\begin{aligned} \text{At } t = 2 \text{ ms:} \\ v(2 \text{ ms}) &= \frac{1}{1 \times 10^{-3}} \left(\overset{(1)}{100 \times 10^{-3} \times 2 \times 10^{-3}} \right) \\ &= \frac{100 \times 10^{-6}}{1 \times 10^{-3}} = 100 \times 10^{-3} \text{ V} = \\ &= \underline{100 \text{ mV}} \end{aligned}$$

$$\begin{aligned} \text{At } t = 5 \text{ ms:} \\ v(5 \text{ ms}) &= \frac{1}{1 \times 10^{-3}} \left(\overset{(1)}{100 \times 10^{-6}} + \overset{(2)}{(3 \times 10^{-3} \times 100 \times 10^{-3})} \right) \\ &= \frac{1}{1 \times 10^{-3}} (400 \times 10^{-6}) = \underline{400 \text{ mV}} \end{aligned}$$

Example 2: Capacitor – energy storage

What is the voltage across a $3\mu\text{F}$ capacitor if the charge on one plate is 0.12 mC . How much energy is stored in the capacitor?

Example 2: Capacitor – energy storage

What is the voltage across a $3\mu\text{F}$ capacitor if the charge on one plate is 0.12 mC . How much energy is stored in the capacitor?

Voltage:

$$q = CV$$

$$\therefore V = \frac{q}{C} = \frac{0.12 \times 10^{-3} \text{ C}}{3 \times 10^{-6} \text{ F}}$$

$$\therefore V = 0.04 \times 10^3 \frac{\text{C}}{\text{F}} = \underline{40 \text{ V}}$$

energy:

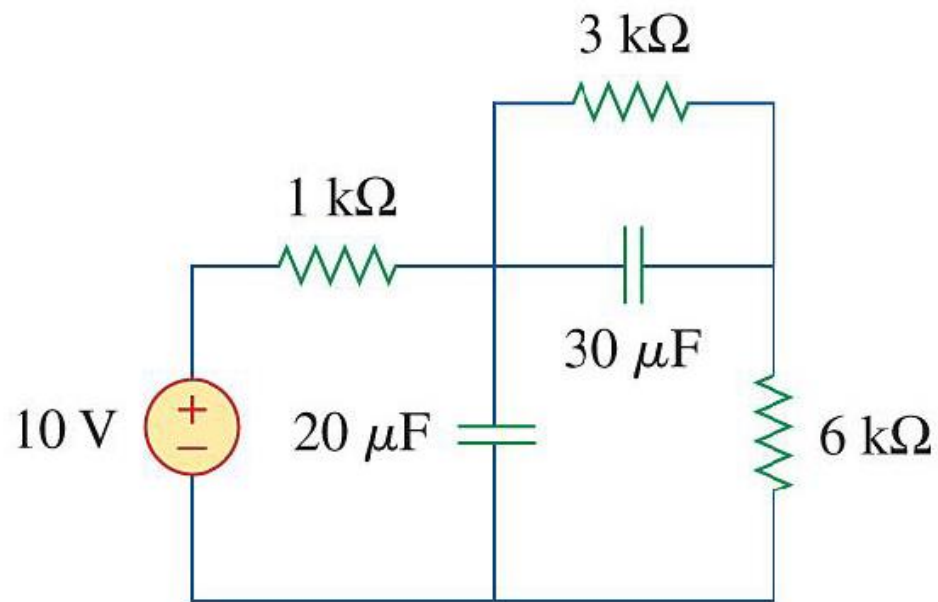
$$W = \frac{1}{2} CV^2$$

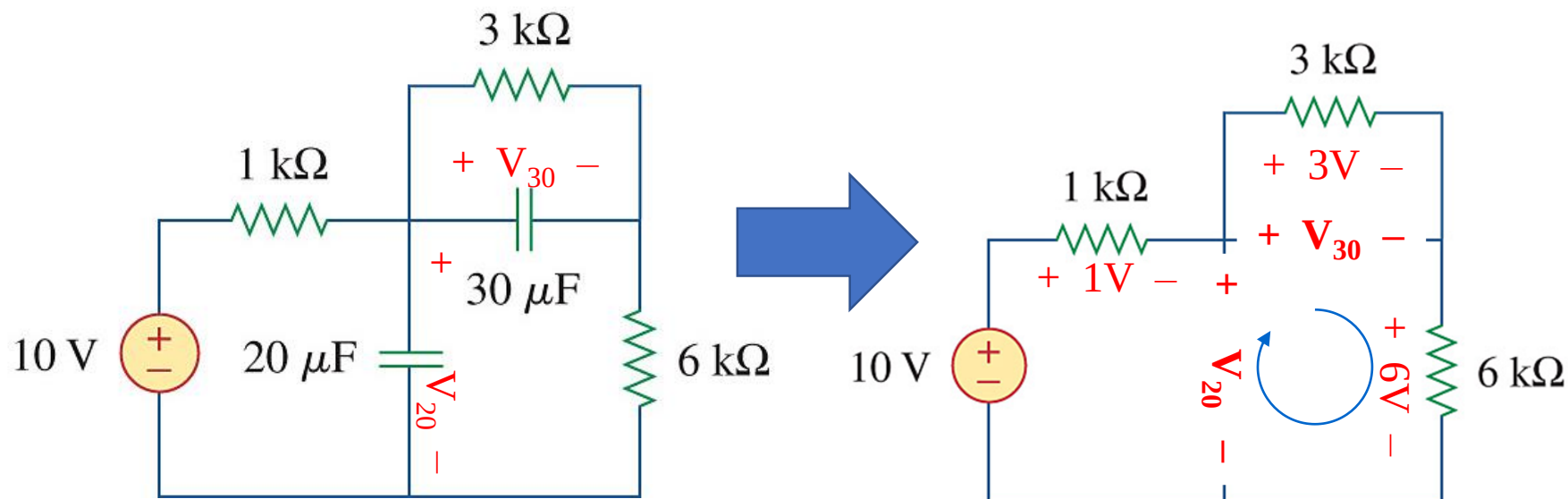
$$= \frac{1}{2} (3 \times 10^{-6}) (40 \text{ V}^2)$$

$$= 2400 \times 10^{-6} = \underline{2.4 \text{ mJ}}$$

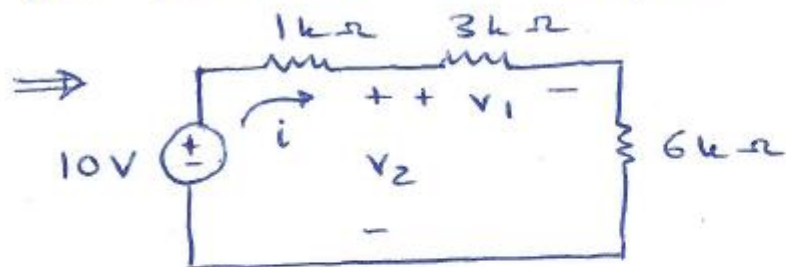
Example 3: Capacitor – energy storage at DC condition

For DC conditions, calculate the energy stored in the two capacitors.





DC conditions $\Rightarrow$ capacitors = open circuits



$$i = \frac{10V}{(1k\Omega + 3k\Omega + 6k\Omega)}$$

$$i = \frac{10V}{10k\Omega} = 1mA$$

V_1 (across 30μF): $V_1 = 3k\Omega (i) = 3k\Omega \times 1mA = 3V$

V_2 (across 20μF): $V_2 = 10 \left(\frac{9k\Omega}{9k\Omega + 1k\Omega} \right)$ voltage division

$$V_2 = 9V$$

Example 3: Capacitor – energy storage at DC condition

For DC conditions, calculate the energy stored in the two capacitors.

$$V_1 \text{ (across } 30\mu\text{F)}: V_1 = 3\text{k}\Omega (i) = 3\text{k}\Omega \times 1\text{mA} = 3\text{V}$$

$$V_2 \text{ (across } 20\mu\text{F)}: V_2 = 10 \left(\frac{9\text{k}\Omega}{9\text{k}\Omega + 1\text{k}\Omega} \right) \text{ voltage division}$$

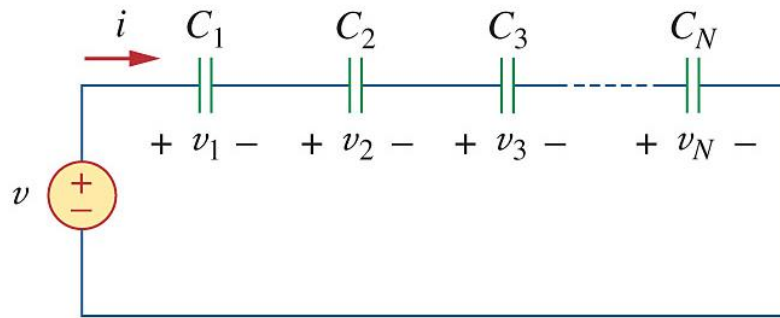
$$V_2 = 9\text{V}$$

$$\begin{aligned} W_1 \text{ (stored in } 30\mu\text{F)}: W_1 &= \frac{1}{2} C_1 V_1^2 \\ &= \frac{1}{2} (30 \times 10^{-6}) (3^2) = \underline{135\mu\text{J}} \end{aligned}$$

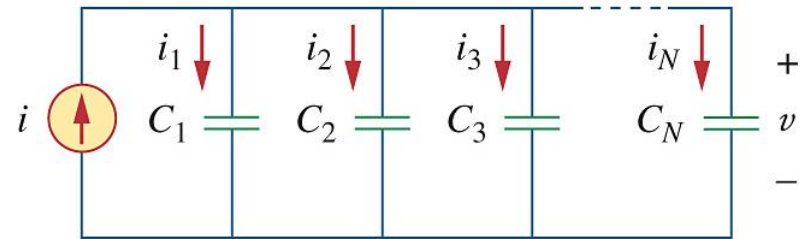
$$\begin{aligned} W_2 \text{ (stored in } 20\mu\text{F)}: W_2 &= \frac{1}{2} C_2 V_2^2 \\ &= \frac{1}{2} (20 \times 10^{-6}) (9^2) = \underline{810\mu\text{J}} \end{aligned}$$

Series and parallel capacitors

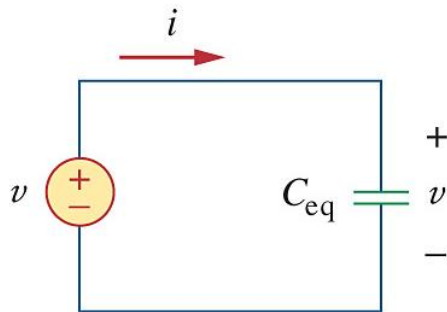
- Series and parallel combinations simplify circuits
- Equivalent capacitance C_{eq} can be used



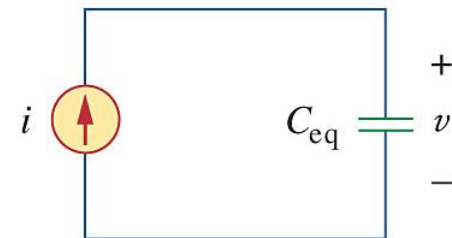
(a)



(a)



(b)



(b)

Series capacitors

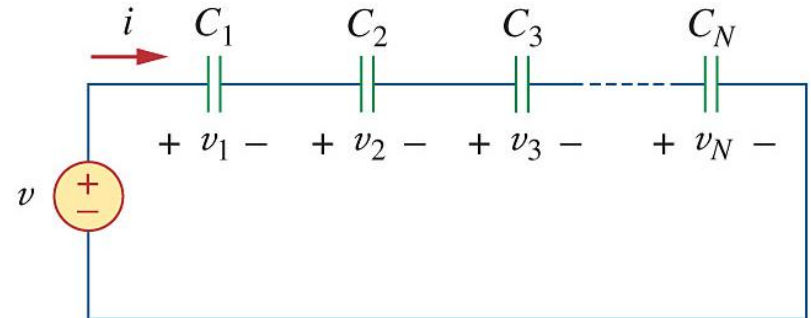
- Current through each capacitor is the same
- Applying KVL:

$$v = v_1 + v_2 + v_3 + \dots + v_N$$

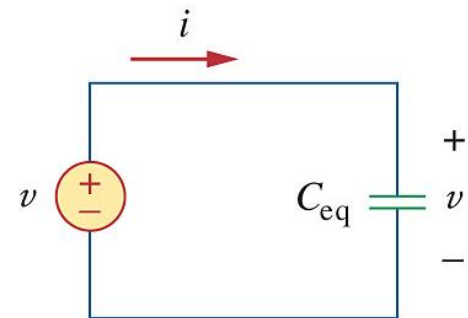
$$v = \frac{1}{C_{eq}} \int_{t_0}^t i(\tau) d\tau + v_1(t_0) =$$

$$\frac{1}{C_1} \int_{t_0}^t i(\tau) d\tau + v_1(t_0) + \dots + \frac{1}{C_N} \int_{t_0}^t i(\tau) d\tau + v_N(t_0)$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$



(a)



(b)

Series capacitors

→ similar to parallel resistors

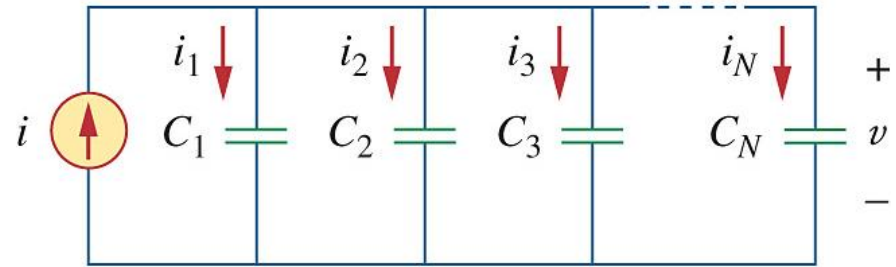
Parallel capacitors

- Voltage across each capacitor is the same
- Applying KCL:

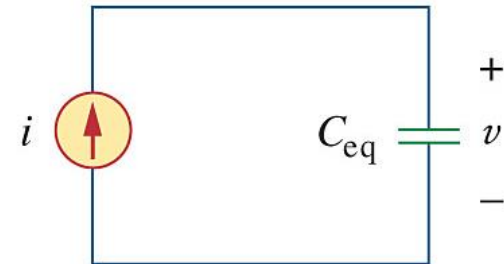
$$i = i_1 + i_2 + i_3 + \cdots + i_N$$

$$i = C_{eq} \frac{dv}{dt} = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + C_3 \frac{dv}{dt} + \cdots + C_N \frac{dv}{dt}$$

$$C_{eq} = C_1 + C_2 + \cdots + C_N$$



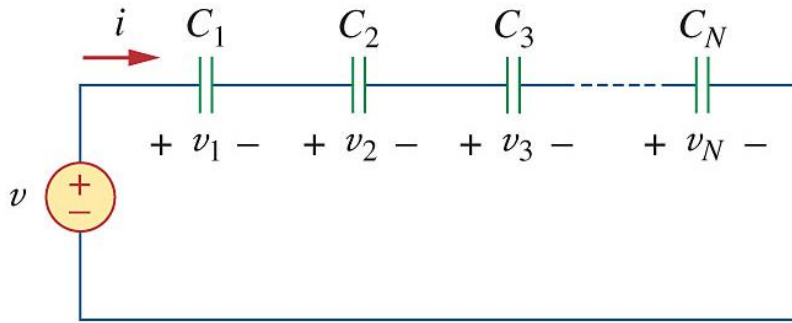
(a)



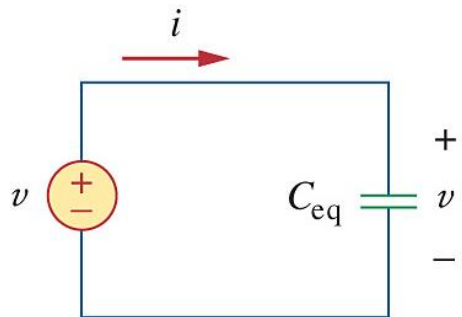
(b)

**Parallel capacitors
= sum of capacitance**

Series and Parallel capacitors

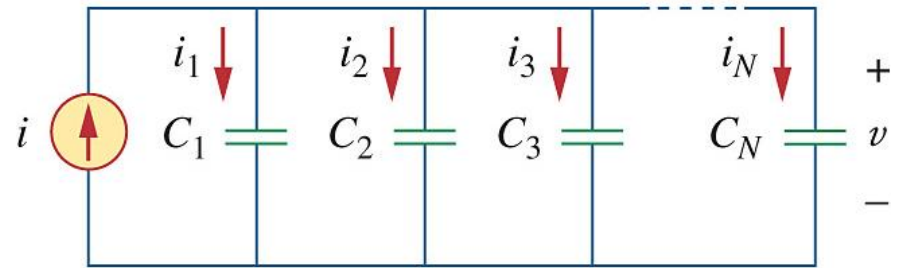


(a)

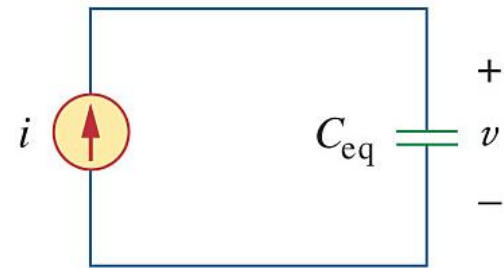


(b)

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}$$



(a)

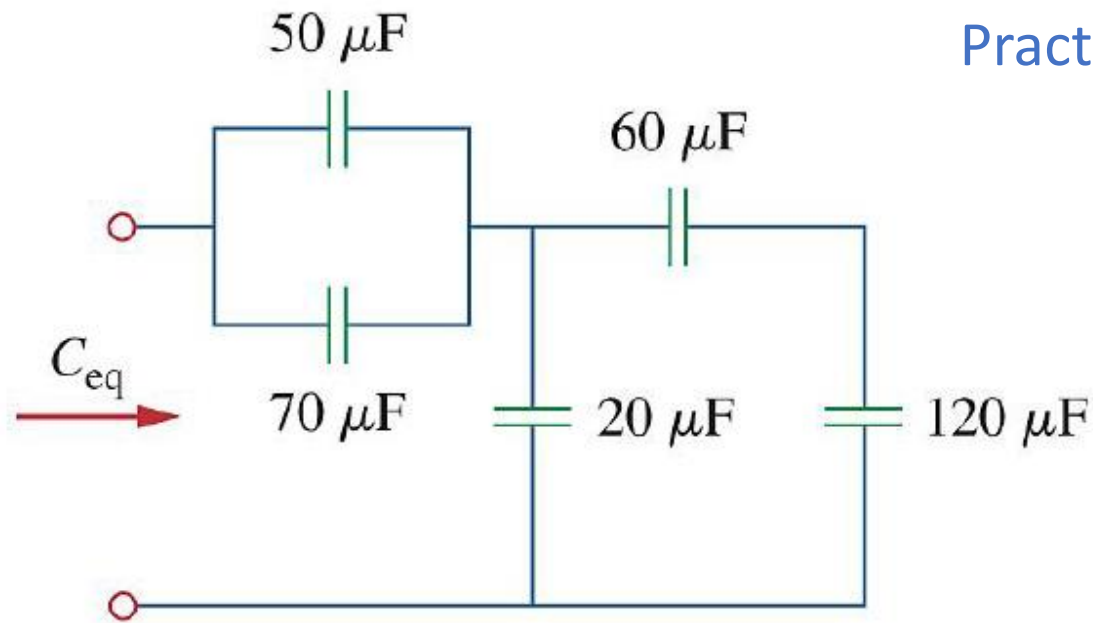


(b)

$$C_{eq} = C_1 + C_2 + \dots + C_N$$

Example 4: Capacitor – equivalent capacitance

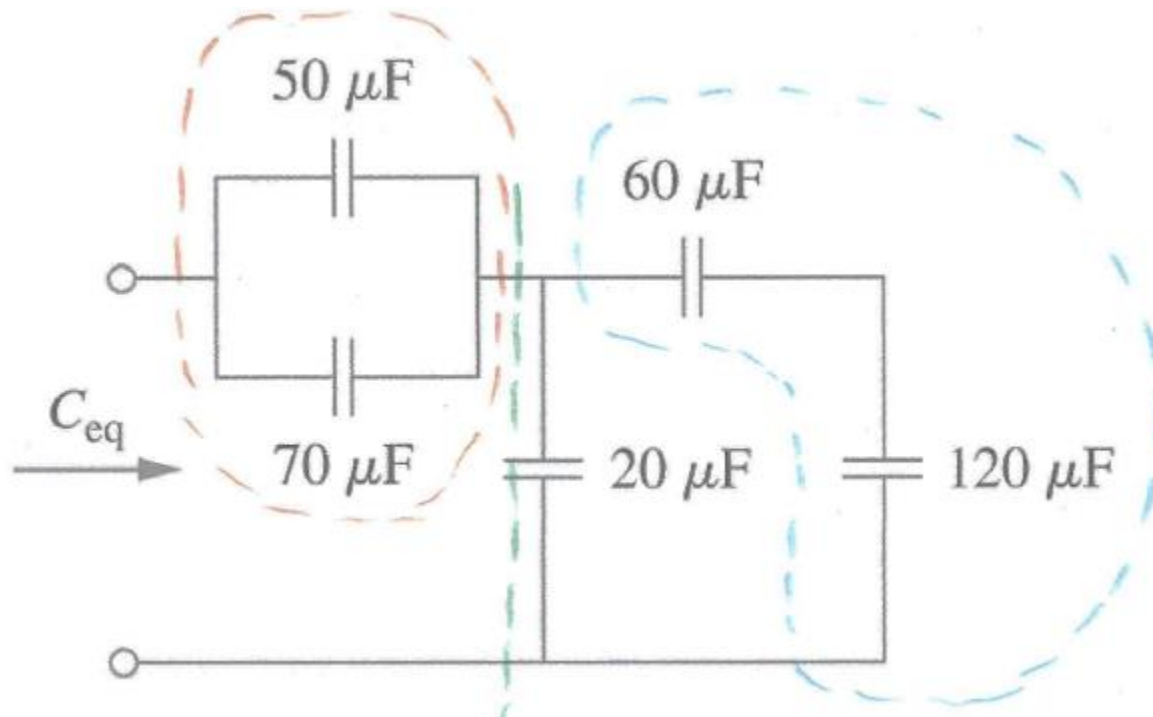
Determine the equivalent capacitance.



Practice Problem 6.6

Example 4: Capacitor – equivalent capacitance

Determine the equivalent capacitance.



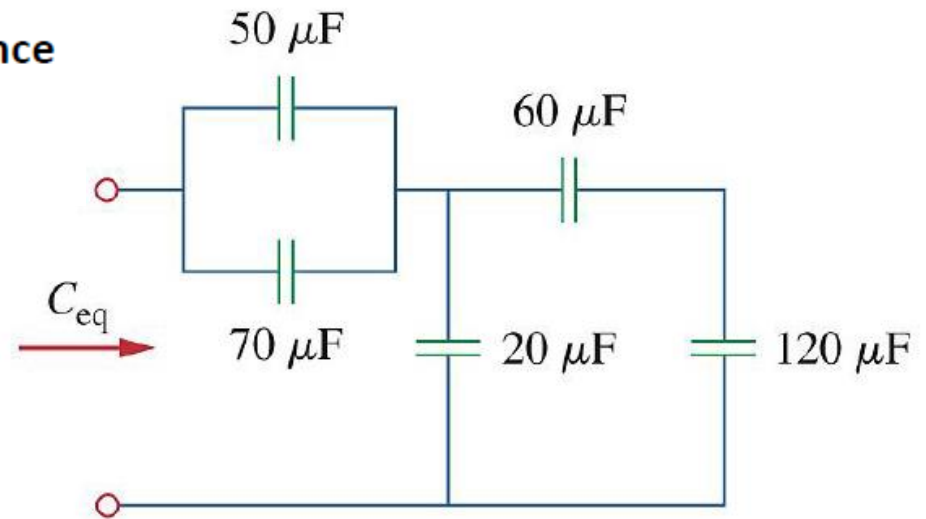
$$\textcircled{*} \quad 60\ \mu\text{F} + 120\ \mu\text{F} = \frac{60 \times 120}{60 + 120} = \frac{7200}{180} = \underline{40\ \mu\text{F}}$$

$$\textcircled{*} \quad 20\ \mu\text{F} \parallel \underline{40\ \mu\text{F}} = 20 + 40 = \underline{60\ \mu\text{F}}$$

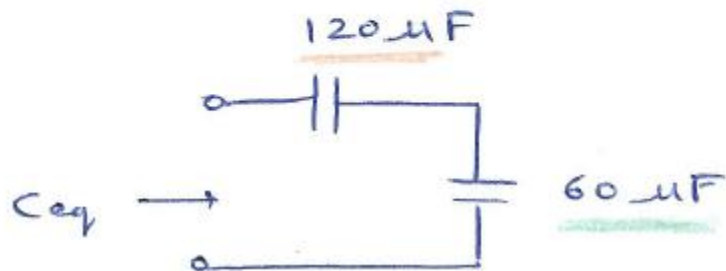
$$\textcircled{*} \quad 50\ \mu\text{F} \parallel 70\ \mu\text{F} = 50 + 70 = \underline{120\ \mu\text{F}}$$

Example 4: Capacitor – equivalent capacitance

Determine the equivalent capacitance.



⊛ $50\ \mu\text{F} \parallel 70\ \mu\text{F} = 50 + 70 = \underline{120\ \mu\text{F}}$



$$C_{eq} = 120\ \mu\text{F} + 60\ \mu\text{F}$$

$$C_{eq} = \frac{120 \times 60}{120 + 60}$$

$$\therefore C_{eq} = \underline{40\ \mu\text{F}}$$